

Algebra 1 — Homework 3

Source: Tutorial sheet 3 (Tutorials/Tutorial 3.pdf) Opened 21 Aug 2026 Due Mon 31 Aug 2026, 12:00 AM

Submit: 2(a), 9(b), 10(a). Optional: 1, 2(b), 4(b), 6, 7 (“if you are not fully confident about any of these, write a full solution”).

Required

2(a). Let A be the coefficient matrix of Tutorial 2 problem 2(a), i.e. $A = \begin{pmatrix} 1 & 2 & 1 & 3 & 2 \\ 2 & 4 & 0 & 4 & 6 \\ 1 & 2 & 2 & 5 & 1 \end{pmatrix}$.

Find with justification a basis for $\text{Nul}(A)$, $\text{Row}(A)$, $\text{Col}(A)$.

9(b). Write $A \sim B$ if one can go from A to B by a combination of row and column operations. Find nice representatives for the equivalence classes, similar to RREF for row equivalence. (“The answer is very simple!”)

10(a). Using only the vector space axioms, prove that the zero vector of a vector space V is unique.

Optional

1. Let R be in REF. (a) Its nonzero rows are independent, hence a basis of $\text{Row}(R)$. (b) Its pivot columns are independent and span $\text{Col}(R)$.

2(b). Having done 1 and 2(a), do 2(a) for an arbitrary matrix A . Explain and justify your procedure for each subspace. Can you find multiple bases for each?

4(b). If you used A^T to find a basis of $\text{Col}(A)$ in problem 2, find another method using only row operations on A .

6. For square A , justify why all 10 statements in the 2×5 table of equivalences (Tutorial 2, problem 5) become equivalent.

7. For an $r \times c$ matrix A with $f(x) = Ax$, show any two of the following imply the third: (i) $r = c$; (ii) f is injective; (iii) f is surjective. (Analogy with functions between finite sets?)